

Quiz (Carolina Reaper)

- Q1.** A shipping company's cost function is given by $C(x) = x^2 - 6x + 9$. Write this function in vertex form.
- (A) $C(x) = (x - 3)^2$
(B) $C(x) = (x + 3)^2$
(C) $C(x) = (x - 2)^2$
(D) $C(x) = (x + 2)^2$
(E) $C(x) = (x - 1)^2$
- Q2.** A traffic flow model is represented by the equation $y = x^2 - 4x - 5$. Write this equation in intercept form.
- (A) $y = (x - 5)(x + 1)$
(B) $y = (x + 5)(x - 1)$
(C) $y = (x - 2)(x + 3)$
(D) $y = (x + 2)(x - 3)$
(E) $y = (x + 1)(x - 5)$
- Q3.** The delivery time t (in hours) for a package is given by $t(x) = x^2 + 2x + 1$, where x is the distance (in miles). Identify the vertex of this parabola.
- (A) (0, 1)
(B) (1, 0)
(C) (-1, 0)
(D) (0, 0)
(E) (-1, 1)
- Q4.** A logistics company's profit function is $P(x) = -x^2 + 10x - 25$. Determine the x -intercepts of this function.
- (A) (5, 0) and (-5, 0)
(B) (0, 0) and (10, 0)
(C) (5, 0) and (0, 0)
(D) (0, 0) and (25, 0)
(E) (10, 0) and (0, 0)
- Q5.** The cost C (in dollars) of shipping x packages is given by $C(x) = x^2 - 12x + 36$. Write this function in vertex form and identify the vertex.
- (A) $C(x) = (x - 6)^2$, vertex: (6, 0)
(B) $C(x) = (x - 6)^2$, vertex: (6, 36)
(C) $C(x) = (x + 6)^2$, vertex: (-6, 0)
(D) $C(x) = (x + 6)^2$, vertex: (-6, 36)
(E) $C(x) = (x - 12)^2$, vertex: (12, 0)
- Q6.** A package delivery company's revenue function is $R(x) = -2x^2 + 20x - 50$. Determine the y -intercept of this function.
- (A) (0, -50)
(B) (0, 50)
(C) (0, 0)
(D) (0, 25)
(E) (0, 100)
- Q7.** The distance d (in miles) a driver travels in t hours is given by $d(t) = t^2 + 4t + 4$. Write this equation in intercept form.
- (A) $d(t) = (t + 2)(t + 2)$
(B) $d(t) = (t - 2)(t - 2)$
(C) $d(t) = (t + 4)(t + 1)$
(D) $d(t) = (t - 4)(t + 1)$
(E) $d(t) = (t + 1)(t + 4)$
- Q8.** A shipping company's profit function is $P(x) = x^2 - 8x + 16$. Identify the x -coordinates of the x -intercepts.
- (A) 0 and 16
(B) 4 and 4
(C) 0 and 8
(D) 8 and 0
(E) 4 and 0

Open Ended Questions (FRQ)

- Q9.** Suppose the cost C (in dollars) of transporting x tons of cargo is given by the quadratic function $C(x) = ax^2 + bx + c$. If the vertex of this parabola is $(-3, 100)$, and the y -intercept is $(0, 150)$, find the values of a , b , and c . Show your work.
- Q10.** A logistics company uses the quadratic function $t(x) = ax^2 + bx + c$ to model the time t (in hours) it takes to transport x miles of cargo. If the x -intercepts are $(0, 0)$ and $(10, 0)$, and the vertex is $(5, 2)$, write this function in both vertex and intercept form. Show your work.

Chef's Tips

- Chef's Tip 1: Ensure students use the correct formula for vertex and intercept form.
- Chef's Tip 2: Remind students to identify and interpret the parameters in each form.
- Chef's Tip 3: Encourage students to check their work and provide clear explanations for FRQs.